

Omar Aguilar

346-593-6154 | omaraguilarwork97@gmail.com | [LinkedIn](#) | [GitHub](#)

EDUCATION

University of Houston

Expected May 2028

Bachelor of Science in Computer Science | GPA: 4.0 | Houston, TX

Lone Star College

December 2025

Associate of Arts, Computer Science Pathway | GPA: 3.67 | Houston, TX

PROJECT EXPERIENCE

Persistent Task Tracker

August 2026

JavaScript, HTML, CSS

- Built a JavaScript task manager with 3 filter views (All, Active, Completed), a remaining-task counter, and empty-input validation to keep task lists organized.
- Persisted arrays of task objects with localStorage and JSON, restoring text and completion state across sessions while filtering invalid records and handling malformed saved JSON.
- Implemented inline editing with Enter-to-save and Escape-to-cancel controls, plus accessible labels and live status announcements for task actions.

K-Nearest Neighbors Bakery Demand Predictor

April 2026

C++ | Course Project

- Built a K-nearest neighbors demand predictor using a 20-day bakery dataset and $k = 4$, with weather, weekend/holiday status, and game-day status as input features.
- Calculated Euclidean distances and ranked neighboring records to estimate daily bakery demand, producing a sample prediction of approximately 70.5 loaves.

C++ Raycaster Engine

January 2025 - February 2025

C++, SFML | Personal Project

- Developed a pseudo-3D raycasting engine with 3 core systems for wall detection, distance calculations, and frame rendering.
- Enabled real-time movement and rotation with keyboard controls, adding distance-based wall shading to provide depth cues in the first-person scene.

WORK EXPERIENCE

Digital Personal Shopper

January 2026 - August 2026

Walmart

- Handled an estimated 100+ online orders on busy shifts, coordinating fulfillment, substitutions, staging, and customer pickups through handheld order-management systems.

ACTIVITIES

CodePathways Peer Mentor | Code[Coogs] x Cougarrettes

September 2026 - Present

- Selected for the Fall 2026 peer mentorship program with weekly 30+ minute 1-on-1 meetings focused on a UH student's academic, technical, and professional development.

UH Computer Science Student Organizations | Member

University of Houston

- Active member of 4 student organizations - Cougar CS, Code Coogs, SACNAS, and MAES - participating in technical, professional-development, and community events.

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, HTML, CSS

Web Technologies: React, Node.js, DOM Manipulation, localStorage, JSON

Tools & Libraries: Git, GitHub, Linux, VS Code, Cursor, Codex, SFML

Concepts: Object-Oriented Programming, Data Structures & Algorithms, Debugging, Version Control